

Symmetry-Breaking-Based Fault Localization in Dual Active Bridge Converters at the Unity-Gain Boundary

Abstract—Dual Active Bridge (DAB) converters are critical for high-power applications due to their bidirectional flow and galvanic isolation. However, frequency-domain diagnostic methods for open-circuit faults (OCFs) suffer from a mathematical singularity at the unity-gain boundary ($k = 1$), where primary and secondary side fault signatures become indistinguishable. This paper proposes an enhanced localization strategy based on transient symmetry breaking. When a fault is detected near $k \approx 1$, a controlled adjustment of the voltage ratio and phase shift is introduced to separate overlapping harmonic signatures, restoring fault discriminability without additional sensors or power interruption. Validation in a 7 kW system shows that the proposed method expands the discriminability margin from 0.01 to 0.15 and achieves accurate identification with a latency of approximately 4 ms. The approach provides a cost-effective, software-based solution to improve DAB converter reliability across the full operating range.

Index Terms—dual active bridge, fault localization, open-circuit fault, frequency-domain analysis, unity-gain boundary, output voltage ripple.

I. INTRODUCTION

The dual active bridge (DAB) converter is widely used in high-power applications due to its bidirectional power flow capability and galvanic isolation [1], [2]. However, the large number of semiconductor switches increases the likelihood of faults, particularly open-circuit faults (OCFs), which can degrade system performance and reliability [3]–[6].

Time-domain methods detect OCFs using instantaneous signals such as current or voltage. They offer fast response but are sensitive to noise and load transients [7]–[9]. To overcome these limitations, frequency-domain methods extract harmonic features from the output voltage ripple, enabling robust and sensor-minimal fault localization [10], [11].

However, a critical limitation arises at the unity-gain condition ($k = 1$, $V_{in} = V_{out}$). At this point, the spectral signatures of primary and secondary faults become indistinguishable [11]. This leads to a collapse of localization capability. As a result, existing approaches often revert to time-domain methods, which increase hardware complexity [12]. A purely frequency-domain solution to resolve this ambiguity remains lacking.

To address this challenge, this paper proposes a symmetry-breaking localization strategy based on transient phase-shift perturbation. The method introduces a controlled shift in the operating point to separate overlapping harmonic signatures and restore discriminability. It relies solely on output voltage ripple and does not require additional sensors. As a result, it enables accurate fault localization under unity-gain conditions.

II. THEORETICAL ANALYSIS OF DAB CONVERTER UNDER OPEN-CIRCUIT FAULTS

A. Principle of DAB Operation and SPS Modulation

The DAB converter employs two active H-bridges and a high-frequency transformer for bidirectional power transfer [13], [14]. Under single phase shift (SPS) modulation, the transferred power is given by

$$P = \frac{NV_{in}V_{out}\phi(\pi - |\phi|)}{2\pi^2 F_s L}, \quad (1)$$

where N is the transformer turns ratio, F_s is the switching frequency, and L is the series inductance. The phase shift is normalized as $d = \phi/\pi$, and the operating point is defined by the voltage ratio:

$$k = \frac{NV_{out}}{V_{in}}. \quad (2)$$

B. Harmonic Characteristic under Open-Circuit Faults (OCF)

Under normal operation, the output voltage ripple is dominated by even-order harmonics. An OCF disrupts this symmetry and introduces a fundamental component [10], [11]. The fault is localized by extracting the fundamental harmonic coefficient w and comparing it with theoretical references

$$w_1 = -1 + 4d + k, \quad (3)$$

$$w_2 = 1 + 4d - k. \quad (4)$$

C. Unity-Gain Singularity Analysis

At the unity-gain condition ($V_{in} = V_{out}$, $k = 1$), the substitution of $k = 1$ into (3) and (4) yields

$$w_1 = 4d, \quad w_2 = 4d. \quad (5)$$

Thus, the reference signatures become identical ($w_1 = w_2$), which eliminates the discriminability gap required for localization. As a result, the diagnostic algorithm cannot distinguish between primary and secondary faults, and localization fails.

III. PROPOSED FAULT LOCALIZATION METHOD

The proposed method addresses the fundamental localization ambiguity encountered at the unity-gain boundary ($k = 1$). While standard frequency-domain localization works effectively in asymmetric regions, the overlap of bridge-side signatures at $k = 1$ necessitates a specialized intervention. The strategy introduces a symmetry-breaking shift to restore the discriminability margin using exclusively the output voltage V_{out} as the diagnostic feature.

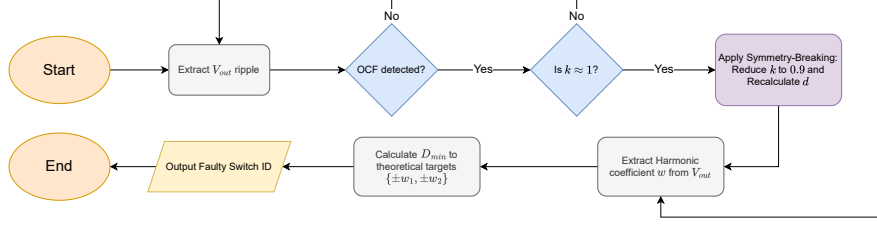


Fig. 1: Flowchart of the integrated diagnostic algorithm.

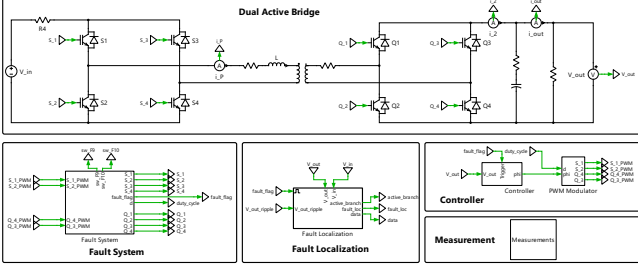


Fig. 2: DAB simulation schematic in PLECS.

TABLE I: System parameters for the DAB converter simulation.

Parameter	Value
Input voltage (V_{in})	780 V
Output voltage (V_{out})	780 V
Rated power (P)	7 kW
Series inductance (L)	507 μ H
Output capacitance (C)	100 μ F
Switching frequency (f_s)	10 kHz
Voltage gain (k)	0.4–1
Phase-shift ratio (d)	0–0.5

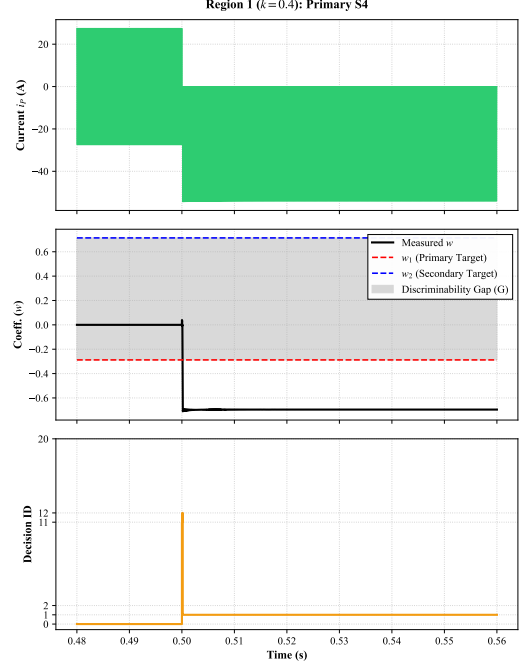


Fig. 3: Localization results for primary-side OCF (S_4) in Region 1 ($k = 0.4$).

A. The Symmetry-Breaking Strategy

The core of the proposed method is the intentional modulation of the voltage conversion ratio k upon fault detection. As defined in (1), the output power is a function of both d and V_{out} . Since k is directly proportional to V_{out} , any intentional shift in the voltage ratio to break the mathematical symmetry will alter the output voltage levels.

To ensure the diagnostic phase does not disrupt the power delivered to the load, the phase-shift ratio d must be dynamically recalculated to an adjusted value d_{adj} to maintain the power reference P . This coordinated transition allows the theoretical targets w_1 and w_2 to diverge, restoring the discriminability gap G while maintaining a stable power reference.

TABLE II: Test matrix for fault localization performance.

Case	Target Region	k Value	d Value
1	1st Region (Buck)	0.4	0.1
2	2nd Region (Buck)	0.8	0.5
3	$k = 1$ Boundary	1.0	0.5
4	Symmetry-Breaking	$1.0 \rightarrow k_{adj}$	$0.15 \rightarrow d_{adj}$

B. Integrated Diagnostic Algorithm

The localization algorithm, illustrated in Fig. 1, monitors the output voltage ripple to detect OCF events and evaluates whether $k \approx 1$. If so, a symmetry-breaking shift is applied by adjusting the voltage ratio and phase shift. The harmonic coefficient is then extracted and compared with updated theoretical targets to identify the faulty switch.

IV. IMPLEMENTATION AND SIMULATION METHODOLOGY

This section details the simulation framework and the specific test scenarios used to validate the proposed localization algorithm.

A. Simulation Environment and System Parameters

The symmetry-breaking strategy was verified using a 7 kW DAB converter model developed in the PLECS simulation environment, as illustrated in Fig. 2. The key electrical parameters and operating ranges utilized for validation are summarized in Table I.

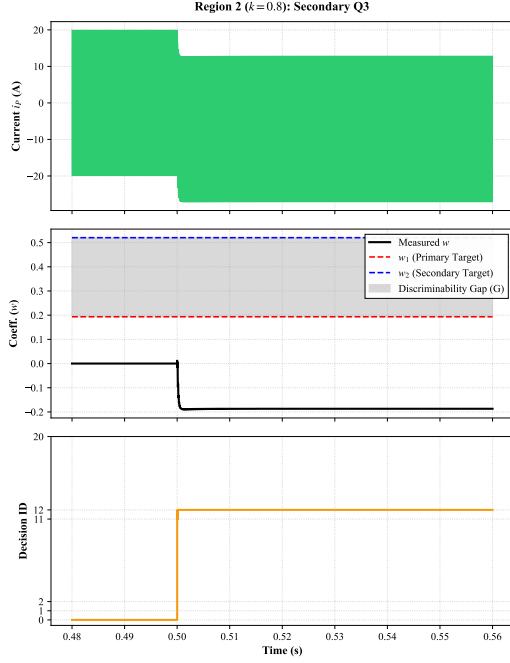


Fig. 4: Localization results for secondary-side OCF (Q_3) in Region 2 ($k = 0.8$).

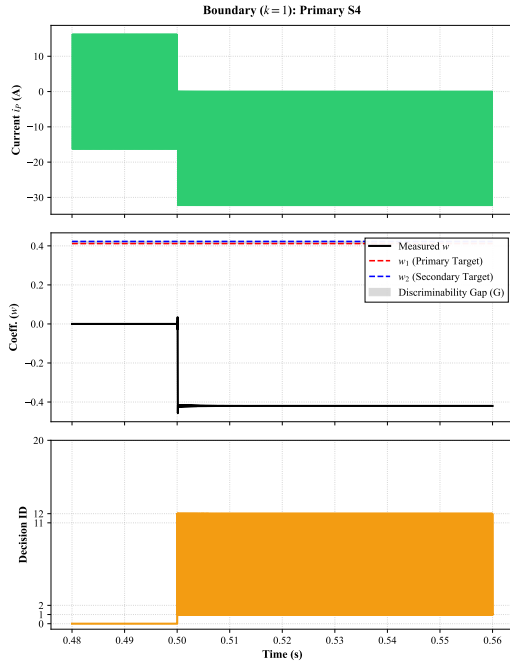


Fig. 5: Localization failure for primary-side OCF (S_4) at the unity-gain boundary ($k = 1$).

B. Fault Generation and Subsystem Logic

OCFs are emulated by suppressing the gate signals of specific diagonal switch pairs after the system reaches steady state at $t = 0.5$ s. Both primary- and secondary-side faults are investigated to assess cross-bridge discriminability.

The fault localization subsystem processes the extracted output voltage ripple using a Fast Fourier Transform (FFT) to determine the fundamental harmonic magnitude w . This

measured value is subsequently used to calculate the adjusted theoretical parameters $w_{1_{adj}}$ and $w_{2_{adj}}$ based on (3) and (4). The localization decision is finalized by comparing the sensed operating variables (V_{in} , V_{out} , and d) against these dynamically updated targets.

C. Evaluation Methodology for Variable k Conditions

The reliability of the algorithm was evaluated across the entire operating range using the test matrix detailed in Table II:

- 1) **Non-Unity Gain Evaluation ($k \neq 1$):** Testing localization performance in both region 1 ($k = 0.4$) and region 2 ($k = 0.8$) to confirm inherent signature diversity.
- 2) **Unity Gain Evaluation ($k = 1$):** Operating at the boundary to demonstrate the failure of standard frequency-domain methods due to signature convergence.
- 3) **Symmetry-Breaking Verification:** Activating the transient perturbation strategy at $k = 1$ to evaluate recovery speed and identification accuracy.

V. RESULTS AND ANALYSIS

A. Performance under Non-Unity Gain ($k \neq 1$)

The baseline method is evaluated under non-unity gain conditions, specifically region 1 ($k = 0.4$) and region 2 ($k = 0.8$). The numerical results for these scenarios are detailed in Table III.

The results demonstrate that the feature extraction provides high numerical convergence, with the residual error (D_{min}) consistently below 0.02. In region 1, the gap is observed to be approximately 1.00, providing high immunity to signal noise. As the converter approaches a higher voltage ratio in region 2 ($k = 0.8$), the gap narrows to approximately 0.31. Despite this reduction, the margin remains an order of

TABLE III: Fault Localization performance in non-unity gain regions ($k \neq 1$).

Scenario	Switch	Measured w	Target w	D_{min}	Gap (G)	Status
<i>Region 1 ($k = 0.4$)</i>						
S_4 (Pri)		-0.6956	-0.7137	0.0181	1.0015	Success
S_3 (Pri)		0.6956	0.7137	0.0182	1.0018	Success
Q_4 (Sec)		0.2771	0.2935	0.0165	1.0082	Success
Q_3 (Sec)		-0.2771	-0.2934	0.0164	1.0081	Success
<i>Region 2 ($k = 0.8$)</i>						
S_4 (Pri)		-0.5031	-0.5196	0.0165	0.3077	Success
S_3 (Pri)		0.5031	0.5196	0.0165	0.3078	Success
Q_4 (Sec)		0.1869	0.1933	0.0064	0.3270	Success
Q_3 (Sec)		-0.1869	-0.1934	0.0065	0.3269	Success

TABLE IV: Fault localization performance at the unity-gain boundary ($k = 1$).

Switch	Measured w	Target w	D_{min}	Gap (G)	Status
S_4 (Pri)	-0.4200	-0.4227	0.0027	0.0109	Failed
S_3 (Pri)	0.4200	0.4227	0.0027	0.0110	Failed
Q_4 (Sec)	0.3914	0.4025	0.0111	0.0203	Failed
Q_3 (Sec)	-0.3914	-0.4026	0.0112	0.0202	Failed

TABLE V: Impact of symmetry breaking on discriminability margin ($k : 1.0 \rightarrow 0.9$).

Switch	Pre-shift G	Post-shift G	Measured w	Target w	D_{min}	Status
S_4 (Pri)	0.0109	0.1526	-0.4141	-0.4298	0.0157	Success
S_3 (Pri)	0.0110	0.1527	0.4141	0.4298	0.0158	Success
Q_4 (Sec)	0.0203	0.1757	0.2471	0.2563	0.0093	Success
Q_3 (Sec)	0.0202	0.1756	-0.2471	-0.2564	0.0093	Success

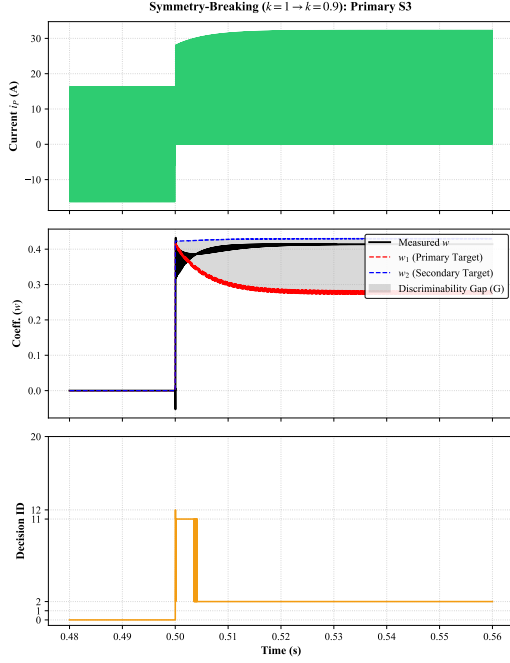


Fig. 6: Localization recovery via symmetry-breaking ($k: 1 \rightarrow 0.9$) for primary-side OCF (S_3).

magnitude larger than the observed residual error, ensuring the algorithm works across both bridges.

The dynamic response for $k \neq 1$ is illustrated in Fig. 3 and Fig. 4. In both cases, the primary current i_P exhibits immediate asymmetry upon fault injection at $t = 0.5$ s. As shown in Fig. 3, the wide discriminability gap allows the harmonic coefficient w to settle rapidly. Similarly, Fig. 4 confirms that even with the reduced margin at $k = 0.8$, the algorithm successfully identifies the secondary-side fault (Q_3) within approximately one switching cycle.

B. Localization Failure at the Unity-Gain Boundary ($k = 1$)

The fundamental limitation of ripple-based localization occurs at $k = 1$, as shown in Table IV. Although the residual error remains low ($D_{\min} \approx 0.0027$), the functional localization fails because the primary (w_1) and secondary (w_2) targets become mathematically indistinguishable.

As visually explained in Fig. 5, the extracted feature w follows a trajectory that the algorithm erroneously maps to the opposite bridge. For a primary-side fault on S_4 , the decision ID stabilizes on a secondary-side identifier. This cross-bridge misidentification occurs because the discriminability gap (G) collapses to less than 0.02, removing the safety margin required for reliable discrimination.

C. Validation of the Proposed Symmetry-Breaking Strategy

To resolve the ambiguity at $k = 1$, the converter shifts its voltage ratio to $k = 0.9$. As shown in Table V, this shift expands the gap G from 0.01 to approximately 0.15, ensuring signatures no longer overlap.

The dynamic recovery is captured in Fig. 6. At $t = 0.5$ s, the decision ID momentarily identifies a secondary-side fault (ID 11) due to target overlap. As the k -shift is implemented,

the widening discriminability gap drives the feature w toward the correct target. The logic self-corrects to the true faulty switch (S_3 , ID 2) within approximately 4 ms.

D. Comparative Discussion and Limitations

The proposed method provides significant advantages in reliability and complexity. In the baseline method, the discriminability gap G collapses by over 98% at $k = 1$, while the symmetry-breaking strategy restores it to 0.15 or greater. This software-only solution utilizes existing control loops without additional hardware probes. While a transient 10% drop in output voltage occurs during the shift, the diagnostic latency is minimal (< 10 ms), allowing for rapid fault mitigation.

VI. CONCLUSION

This paper proposed a symmetry-breaking strategy to resolve the localization failure of frequency-domain methods at the unity-gain boundary ($k = 1$) in DAB converters. By introducing a transient perturbation in the voltage ratio and phase shift, the method restores discriminability between primary and secondary fault signatures.

PLECS results confirm that the proposed approach increases the discriminability margin from 0.01 to 0.15 and enables accurate fault localization across the full operating range with a latency of approximately 4 ms. Although a brief voltage deviation is introduced, the fast response ensures effective fault identification and enables timely mitigation.

The proposed method provides a low-cost, software-based solution that overcomes the unity-gain limitation and improves the reliability of DAB converter fault diagnosis.